XRPL Validator — Daily Engineering Report · 2026-10-08

Period: 2026-10-08 00:00 to 24:00 US Eastern (2026-10-08 04:00 to 2026-10-09 04:00 UTC); ledgers #107,508,606 to #107,530,919. Every count from the ledger and the validator's logs stops at #107,530,919, closed 03:59:52 UTC. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores. Not available: Steps 2–5 receipt records (they begin 2026-10-09).

Status

Degraded (all causes resolved or fixed) — coverage outside planned maintenance 99.166% (objective 99.9%): 186 ledgers unsigned.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,120 of the 22,314 ledgers the network closed (99.131%); every validation matched the network's validated ledger. Unsigned: 8 during planned restarts, 186 outside maintenance (1 incident).
- **Native engine:** 19 engine receipts recorded (2 fixed, 17 explained, 0 without a recorded fix).
- **Changes:** 1 deployment (cycle 166), 7 commits merged into main.
- **Network:** 1 slow-close period; the longest, 15:10 UTC–15:10 UTC, closed ledgers up to 17 s apart.
- **Amendment fixBatchV1_2** holds majority; activation due ~2026-10-09 14:12 UTC.
- **Amendment BatchV1_1** holds majority; activation due ~2026-10-09 14:46 UTC.

Service levels

Measure	Value
Ledgers closed by the network	22,314
Validations issued	22,120
Coverage (validations / ledgers closed)	99.131%
Validations matching the network's validated ledger hash	22,120 of 22,120

Measure	Value
Unsigned during planned restarts	8
Unsigned outside maintenance	186
Coverage outside planned maintenance (objective 99.9%)	99.166%
Restarts	2
Basis: the hash the first peer announced as accepted (before cycle 183)	22,120
External agreement, data.xrpl.org (UTC day 2026-10-08)	0.98945 (216 missed of 20,481)

Restarts:

- 21:46:48 UTC: restart (no deployment recorded); validations resumed at #107,525,189 (186 ledgers unsigned between the last validation before the restart and the first after it)
- 00:14:32 UTC: deployment of cycle 166; validations resumed at #107,527,426 (8 ledgers unsigned between the last validation before the restart and the first after it)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Earlier FFI shadow log (libxrpl applied against the validator's mirror; superseded by Step 1): 33 transaction-level differences recorded this day, across 17 ledgers. These were not triaged into receipts.

Ledgers with a receipt in the period (#107,508,606–#107,530,919), by step:

Step	Ledgers with a receipt
Step 1 · native engine	19
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	21:30 UTC	#107,524,867	Step 1 · native engine	4 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-02	21:35 UTC	#107,524,947	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 ledger change(s) the network made were absent from ours: 1851FA6B	fixed: commit 4adb4a8, deployed in cycle 166 (2026-10-09 00:15 UTC)

ID	Time	Ledger(s)	Step	Description	Disposition
R-03	21:35 UTC	#107,524,956	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 change(s) of ours the network did not make	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-04	21:35 UTC	#107,524,958	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 ledger change(s) the network made were absent from ours: 1851FA6B	explained: all 1 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-05	21:36 UTC	#107,524,974	Step 1 · native engine	8 change(s) of ours the network did not make	explained: all 2 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-06	21:37 UTC	#107,524,976	Step 1 · native engine	8 change(s) of ours the network did not make	explained: all 2 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-07	21:37 UTC	#107,524,978	Step 1 · native engine	8 change(s) of ours the network did not make	explained: all 2 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-08	21:37 UTC	#107,524,980	Step 1 · native engine	8 change(s) of ours the network did not make	explained: all 2 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-09	21:37 UTC	#107,524,982	Step 1 · native engine	8 change(s) of ours the network did not make	explained: all 2 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-10	21:37 UTC	#107,524,984	Step 1 · native engine	9 change(s) of ours the network did not make	explained: all 2 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-11	21:37 UTC	#107,524,986	Step 1 · native engine	9 change(s) of ours the network did not make	explained: all 2 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-12	21:37 UTC	#107,524,988	Step 1 · native engine	9 change(s) of ours the network did not make	explained: all 2 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-13	21:37 UTC	#107,524,990	Step 1 · native engine	9 change(s) of ours the network did not make	explained: all 2 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-14	21:38 UTC	#107,524,992	Step 1 · native engine	9 change(s) of ours the network did not make	explained: all 2 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-15	21:38 UTC	#107,524,994	Step 1 · native engine	11 change(s) of ours the network did not make	explained: all 2 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-16	21:38 UTC	#107,524,996	Step 1 · native engine	10 change(s) of ours the network did not make	explained: all 2 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-17	21:38 UTC	#107,524,998	Step 1 · native engine	8 change(s) of ours the network did not make	explained: all 2 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-18	21:38 UTC	#107,525,000	Step 1 · native engine	11 change(s) of ours the network did not make	explained: all 2 transactions re-ran byte-exact in an isolated replay (auto-triage)
R-19	21:38 UTC	#107,525,002	Step 1 · native engine	4 change(s) of ours the network did not make; 4 object(s) differed in content	fixed: commit 4adb4a8, deployed in cycle 166 (2026-10-09 00:15 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding,

commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-10-08-01, INC-01 is INC-2026-10-08-01.

Incidents

INC-01 · 186 ledgers unsigned

Window	21:38:41 UTC to 21:50:40 UTC
Ledgers	#107,525,003–#107,525,188 (186)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Cause	The validator halted at #107,525,002. A second DelegateSet that used a Ticket modified an existing Delegate entry; libxrpl 3.4.0 returned temDISABLED, and with the FFI overlay authoritative for the validator's state (Stage 3) that state was wrong. Stage 3 was switched off and the validator was cold-restarted at 21:46 UTC; it was down about 12 minutes. The lasting fix was the native engine writing the state itself.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
166	`4adb4a8`	00:15:21 UTC	1,961 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	410 matched, 0 mismatched	#41 from 00:41 UTC: ended early: superseded by the next deployment (2026-10-09 16:39 UTC)

Cycle 166 delivered: `5c34ef8` chore(license): credit James Turner: MIT with an attribution requirement; `4adb4a8` feat(node): Stage 4 Phase B — the native Rust engine writes state.rocks (XRPL_NATIVE_WRITER=1)

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
23:52 UTC	xrpl-validator	`4adb4a8`	feat(node): Stage 4 Phase B — the native Rust engine writes state.rocks (XRPL_NATIVE_WRITER=1)
00:20 UTC	xrpl-validator	`5b6c81f`	feat(scripts): watch_engine says who writes state.rocks under the native writer
00:51 UTC	xrpl-validator	`d354cf0`	feat(tools): xrpl-watch, the validator's terminal dashboard in Rust (HALCYON)
01:17 UTC	xrpl-validator	`45ca52a`	feat(tools): xrpl-watch shows every transaction type and result code
01:21 UTC	xrpl-validator	`e53ce35`	feat(tools): xrpl-watch lists every type and result on the main screen
01:25 UTC	xrpl-validator	`b945ba2`	feat(tools): xrpl-watch in real time — ~30 fps, the validator read 4x a second
01:46 UTC	xrpl-validator	`c86e5e0`	feat(tools): xrpl-watch --record --fps, and a real-time footer

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **feat(node): Stage 4 Phase B — the native Rust engine writes state.rocks (XRPL_NATIVE_WRITER=1)** (`4adb4a8`, xrpl-validator) On 2026-10-08 the network enabled PermissionDelegationV1_1. libxrpl 3.4.0 refused those transactions (temDISABLED); under Stage 3 its overlay was what got written, so #107525002 failed the state hash three times and the validator halted, while the native engine had applied them exactly as mainnet did.
- **feat(scripts): watch_engine says who writes state.rocks under the native writer** (`5b6c81f`, xrpl-validator) Under XRPL_NATIVE_WRITER the dashboard now has a "WHO WRITES THE LEDGER DATABASE" block: the Rust engine ACTIVELY WRITING (or LOADING, or OFF on the breaker), libxrpl COMPARING ONLY, and xrpld the reference node as the data feed plus the ledgers our engine could not vouch for. The FFI panel is titled COMPARISON ONLY, the native panel ACTIVELY WRITING / LOADING / WRITER OFF, and the writer's share counts ledgers since its copy of …
- **feat(tools): xrpl-watch, the validator's terminal dashboard in Rust (HALCYON)** (`d354cf0`, xrpl-validator) A ratatui successor to scripts/watch_engine.py, run beside the validator: `tools/xrpl-watch`, its own workspace so the validator's build and lock never see its dependencies.
- **feat(tools): xrpl-watch shows every transaction type and result code** (`45ca52a`, xrpl-validator) The C++ panel lists the types and result codes since start, as many as fit in two lines each, ending "+N more · t". The t key opens a page with every type and every result code, counts and log-scaled bars (one tx is still visible next to forty thousand), tes / tec / other coloured apart; t again returns. --view types renders that page with --once and --record.
- **feat(tools): xrpl-watch lists every type and result on the main screen** (`e53ce35`, xrpl-validator) James: take out the event feed (or really reduce it) and the t page. The bottom row is now EVERY TX TYPE and EVERY RESULT CODE since the validator started, packed in columns with a log-scaled magnitude glyph per entry, plus THIS LEDGER; long names are shortened (Permissioned -> Perm, …) before eliding. The event feed is one line, the newest event. The t page, the c key and --view are gone.
- **feat(tools): xrpl-watch in real time — ~30 fps, the validator read 4x a second** (`b945ba2`, xrpl-validator) James: "1 second refresh is fast but can seem kind of choppy". The screen now redraws at ~30 fps and reads the validator's API every 0.25 s (the upstream every 1 s, the process scan every 5 s, amendments every minute, the receipts file only when it changes). Between reads it animates: the counters roll to their new values, the pulse is a real-time trace (each ledger a beat where it closed, scrolling left and fading, …
- **feat(tools): xrpl-watch --record --fps, and a real-time footer** (`c86e5e0`, xrpl-validator) --record takes --fps F: frames F times a second while the data is still read every --interval (the walkthrough video records 15 fps, reads 4x a second). The footer says what the screen now does: real time, ~30 fps, the validator read 4x a second.

Independence program

No independence record for UTC day 2026-10-08; records begin 2026-10-20.

Network

Measure	Value
Ledgers closed	22,314 (#107,508,606–#107,530,919)
Mean close interval	3.87 s
Longest close interval	17 s (before #107,519,003)
Closes at least 15 s apart	1
Empty ledgers	0

- Slow closes 15:10:23 UTC–15:10:40 UTC: ledgers #107,519,003, 1 close at least 15 s apart, up to 17 s.

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Voting events:

- PermissionDelegationV1_1 was enabled — ledger 107,524,865, 2026-10-08 21:29 UTC

Amendment	Majority since	Activation due
fixBatchV1_2	2026-09-25 14:12 UTC	2026-10-09 14:12 UTC
BatchV1_1	2026-09-25 14:46 UTC	2026-10-09 14:46 UTC

Amendments enabled on mainnet: 95.

Open items

- Amendment fixBatchV1_2: activation due ~2026-10-09 14:12 UTC.
- Amendment BatchV1_1: activation due ~2026-10-09 14:46 UTC.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
10-08 22:52	10-08 23:52	xrpl-validator	`4adb4a8`	feat(node): Stage 4 Phase B — the native Rust engine writes state.rocks (XRPL_NATIVE_WRITER=1)
10-09 00:20	10-09 00:20	xrpl-validator	`5b6c81f`	feat(scripts): watch_engine says who writes state.rocks under the native writer

Committed	Merged into main	Repo	Commit	Subject
10-09 00:51	10-09 00:51	xrpl-validator	`d354cf0`	feat(tools): xrpl-watch, the validator's terminal dashboard in Rust (HALCYON)
10-09 01:17	10-09 01:17	xrpl-validator	`45ca52a`	feat(tools): xrpl-watch shows every transaction type and result code
10-09 01:21	10-09 01:21	xrpl-validator	`e53ce35`	feat(tools): xrpl-watch lists every type and result on the main screen
10-09 01:25	10-09 01:25	xrpl-validator	`b945ba2`	feat(tools): xrpl-watch in real time — ~30 fps, the validator read 4x a second
10-09 01:46	10-09 01:46	xrpl-validator	`c86e5e0`	feat(tools): xrpl-watch --record --fps, and a real-time footer

Times in UTC. A commit appears on the day it was committed to main or merged into it.